

Data Engineer Assignment – Satellite Intelligence

Expected effort: 2–3 hours

Deadline: 7 days from when you receive this brief

Deliverable: a GitHub repo + a Loom walkthrough video

Context

At Carnot, we work with messy field data: sensor readings from parcels, satellite-derived vegetation indices, weather feeds, and farm metadata. Production reality is that none of it arrives clean — readings are missing, statuses are inconsistent, joins are messy, and you have to make sensible calls fast.

This take-home is a simplified version of an actual data wrangling task we'd ask you to do on your first week. The goal is to see how you think, not how perfectly you score on any single step.

The data

You'll find two CSVs in [this](#) starter pack:

parcel_readings.csv — daily readings per farm parcel

- parcel_id — string identifier
- date — daily reading date
- ndvi_value — vegetation index value, valid range is [-1, 1]
- temperature_c — daily mean temperature in Celsius
- rainfall_mm — daily rainfall in millimetres
- sensor_status — sensor health flag

parcel_metadata.csv — parcel info

- parcel_id — string identifier
- mill_id — string identifier for the sugar mill this parcel feeds
- crop_type — string (e.g., "sugarcane", "wheat", "soybean")
- sowing_date — date the crop was sown

- `area_hectares` — float
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Your tasks

1. Data quality audit (~30 min)

Read both CSVs and list every data quality issue you find. For each issue:

- What is it?
- How prevalent? (rough count or %)
- Your call: drop / impute / flag / repair — and a 1-sentence justification

Put this in a clear section of your README.

2. Build a clean pipeline (~60 min)

Write a Python script (or notebook) that:

- Ingests both CSVs
- Cleans both per the decisions you made in step 1
- Joins them into a single time-series dataset (one row per `parcel_id` × date)
- Writes the result to `cleaned_parcel_timeseries.csv`

Use any libraries you like — Pandas, NumPy, Polars, whatever you reach for in real work. All coding tools are allowed and expected.

3. Quick analysis (~30 min)

Using your cleaned dataset, for each `crop_type`:

- Compute the mean NDVI in the 30 days before `sowing_date`
- Compute the mean NDVI in the 30 days after `sowing_date`
- Ignore rows where `sensor_status` is bad
- Output a small DataFrame: `crop_type | mean_ndvi_before | mean_ndvi_after | n_parcel`s

Write 2–3 sentences interpreting what you see. Don't tune anything — just produce numbers and explain.

4. Production-readiness reflection (~15 min, in README)

If this pipeline ran daily and the dataset was 100× larger:

- What three things would you change?
- What would you monitor in production?
- What's the most likely thing to silently break?

What to submit

1. A public GitHub repo containing:
 - Your code (script or notebook)
 - Your output `cleaned_parcel_timeseries.csv`
 - A README that includes: your data quality audit, your approach, the analysis output, and the production reflection
2. A Loom walkthrough video (5–10 minutes) in which you:
 - Show your code on screen
 - Narrate why you made the three or four most important decisions
 - Verbally answer the production reflection out loud (not just in the README)

Submit the GitHub link and the Loom link by replying to the email this brief came in.

Allowed

- Any libraries
- AI coding tools (Claude, Cursor, Codex, ChatGPT) — we use them daily here. In your README, briefly note how you used them.
- Web search and documentation
- 7 days of calendar time

Not allowed

- Submitting code without the Loom walkthrough — we won't review code without it. This is the most important rule.
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Evaluation

We are not evaluating:

- Whether you got the "right" answer (there isn't one)
- How fast you finished
- Whether your code is over-engineered (please don't)

If this brief was fun to work on, that's a good sign — for you and for us.